

Rahul Ray

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MSc Data Science student at Leiden with a habit of building things to answer questions I couldn't find clean answers to.
Interested in what actually happens when you push a method past its assumptions.

Education

MSc Computer Science — Data Science

Leiden University, Leiden, Netherlands

Sep 2025 – Aug 2027 (expected)

Bachelor of Technology — Computer Science

SRM Institute of Science and Technology, Chennai, India

2021 – 2025

Work Experience

Machine Learning Intern

Fox Solutions Pvt. Ltd., India

Mar 2025 – Aug 2025 (6 months)

- Developed a computer vision solution using Vision Transformers (ViT), working with state-of-the-art frameworks across the full pipeline from model development through to edge deployment.
- Responsible for adapting ViT architectures to work within the hardware constraints of edge devices, balancing model performance against inference latency and memory footprint.
- Gained hands-on experience shipping ML models into real production environments beyond a research or notebook setting.

Projects

Can LLMs Make Tabular ML Better? — Hybrid Feature Engineering Experiment 🐙 github.com/Rahulrayy

Python · XGBoost · Groq API · LLaMA 3.1 · sentence-transformers · Optuna · SHAP · scipy

- Designed and ran a rigorous experiment testing whether LLM-generated semantic features improve XGBoost classification on UCI Adult Income and the HMEQ Loan Default dataset — the latter specifically chosen to reduce pretraining contamination risk.
- Each row was sent to LLaMA 3.1 8B (via Groq) under three expert framings (career trajectory, financial risk, social capital); outputs parsed into structured fields and encoded as 384-dimensional sentence embeddings via `all-MiniLM-L6-v2`.
- Implemented a full nested CV pipeline (3×5 -fold, 15 outer scores) with **leak-free PCA fitted inside each fold** and Optuna tuning in the inner loop, preventing both embedding leakage and optimistic hyperparameter evaluation.
- The hypothesis was not supported: on Adult the hybrid was statistically indistinguishable from baseline (Cohen's $d = -0.14$); on HMEQ it was significantly worse ($d = -0.84$, $p = 0.004$). Documented the negative result honestly with Wilcoxon signed-rank tests across all prompt framings.
- SHAP analysis confirmed LLM PCA components ranked mid-table — used but not dominant — consistent with noise addition rather than signal enrichment.

AutoInsight — End-to-End AutoML Web Application

🐙 github.com/Rahulrayy

Python · Streamlit · scikit-learn · XGBoost · LightGBM · CatBoost · SHAP · Optuna · Groq API · Plotly

- Built a browser-based AutoML application: users upload a CSV, select a target column, and receive a full ranked model leaderboard, SHAP explainability, and plain-English LLM insights — no code required.
- Trains 7 models in parallel (Logistic Regression, Random Forest, Extra Trees, GBM, XGBoost, LightGBM, CatBoost) with stratified k-fold CV; optional Bayesian tuning via Optuna (40 trials, TPE sampler) on top-ranked models.
- Preprocessing pipeline handles missing value imputation, datetime decomposition, IQR-based outlier capping, one-hot and label encoding, StandardScaler, and SelectKBest feature selection — all configurable via `config.yaml`.
- Deliberate privacy-first design: the LLM receives only *aggregated metrics* (leaderboard scores, SHAP values, profiling summaries), never raw user data, eliminating hallucination risk and avoiding sending sensitive data to external APIs.
- LLM responses stream live (Groq + llama-3.3-70b) with a follow-up Q&A tab; full session exportable as structured JSON.

- Parsed Quake 1’s raw binary BSP v29 format from scratch to extract walkable surfaces and construct navigation graphs across all 38 retail maps, with nodes at face centroids and edges between reachable neighbours.
- Generated ~625,000 source-goal pairs via Dijkstra to compute ground truth path costs; modelled the target as a *correction factor* (true cost / euclidean distance) — a scale-invariant label that generalises across maps of different physical sizes.
- Trained an MLP and an XGBoost regressor on 17 spatial and graph-structural features; plugged each as a multiplicative correction to the euclidean heuristic inside A*.
- **Results:** 42% fewer nodes expanded on average across 38 maps with paths only ~2% longer than optimal. Train/val/test split by episode to prevent map topology leakage into evaluation.
- Both models achieved near-identical reductions (MLP 42.4% vs XGBoost 42.2%), pointing to the hand-crafted feature set as the representation bottleneck — a clean finding with a clear direction for future work.

- Reproduced and studied the epoch-wise double descent phenomenon from Nakkiran et al. (2019) using a CIFAR-adapted ResNet-18 under symmetric label noise ($\eta \in \{0\%, 10\%, 20\%, 30\%\}$), training up to 600 epochs with 3-seed averaging and shaded confidence bands.
- Tracked clean vs. corrupted training accuracy separately at every epoch; the corrupted-accuracy sigmoid inflection aligns closely with the test loss peak — the clearest mechanistic signal, connecting memorisation timing directly to generalisation degradation.
- Tested two regularisation interventions: label smoothing ($\varepsilon = 0.1$) was a clear rejection — final loss 0.87 vs baseline 0.57. Mixup ($\alpha = 0.2$) improved final performance (0.48 vs 0.57) without eliminating the peak, reported honestly as a partial result.
- Entropy analysis showed model confidence stayed elevated through the full memorisation phase and collapsed only during recovery, correcting the original overconfidence hypothesis and yielding a cleaner interpretation.

Technical Skills

Languages	Python, SQL
ML / DL	PyTorch, scikit-learn, XGBoost, LightGBM, CatBoost, SHAP, Optuna, Vision Transformers (ViT)
LLMs / NLP	Groq API, LLaMA 3.x, sentence-transformers, prompt engineering
Tools	Streamlit, pandas, NumPy, Plotly, NetworkX, Git
Concepts	AutoML, nested cross-validation, edge deployment, A* search, BSP parsing, generalisation theory

Languages

English — C1 Professional Proficiency (*IELTS Certified*) | Hindi — Native Bengali — Native

Academic Interests

Tabular learning & LLM integration · Generalisation theory & double descent · Heuristic search & planning · Explainable ML · Edge deployment · JEPA